2 A steel ball is placed on the inside surface of a hollow circular cone. The ball moves in a horizontal circle at constant speed, as shown in Fig. 2.1.

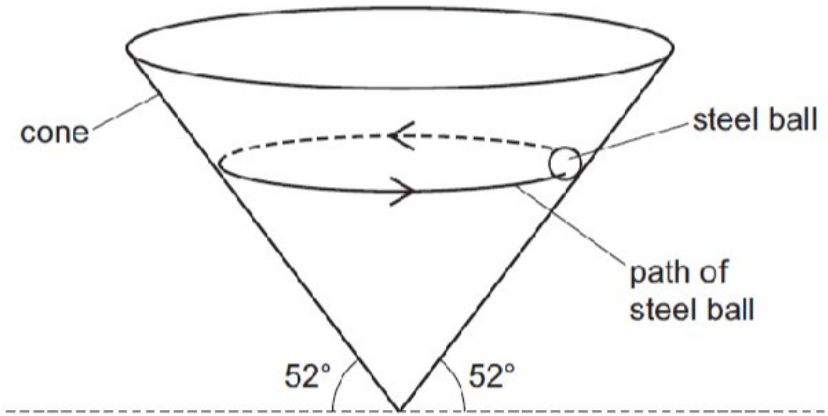


Fig. 2.1

The angle of the side of the cone to the horizontal is 52°. There is no friction between the ball and the cone.

(a) Fig. 2.2 shows a cross-section through the cone and the steel ball.

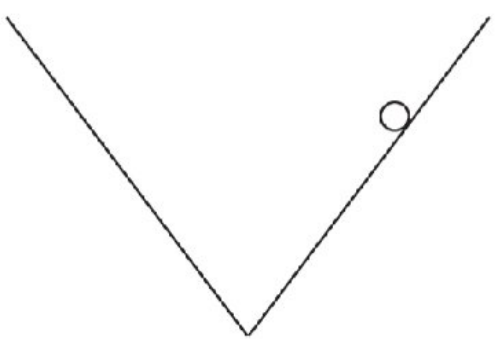


Fig. 2.2

On Fig. 2.2, draw labelled arrows to show the two forces acting on the ball. [2]

(b) Describe how the forces acting on the ball cause its acceleration to be centripetal.

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..... [2]

(c) The ball moves in a circle of radius 0.15 m. Show that the speed of the ball is 1.4 m s^{-1}.

[3]

(d) Calculate the angular speed ω of the ball.

ω = ... rad s^{-1} [1]

(e) The speed of the ball is increased. State and explain what happens to the radius of the circular path of the ball.

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..... [2]